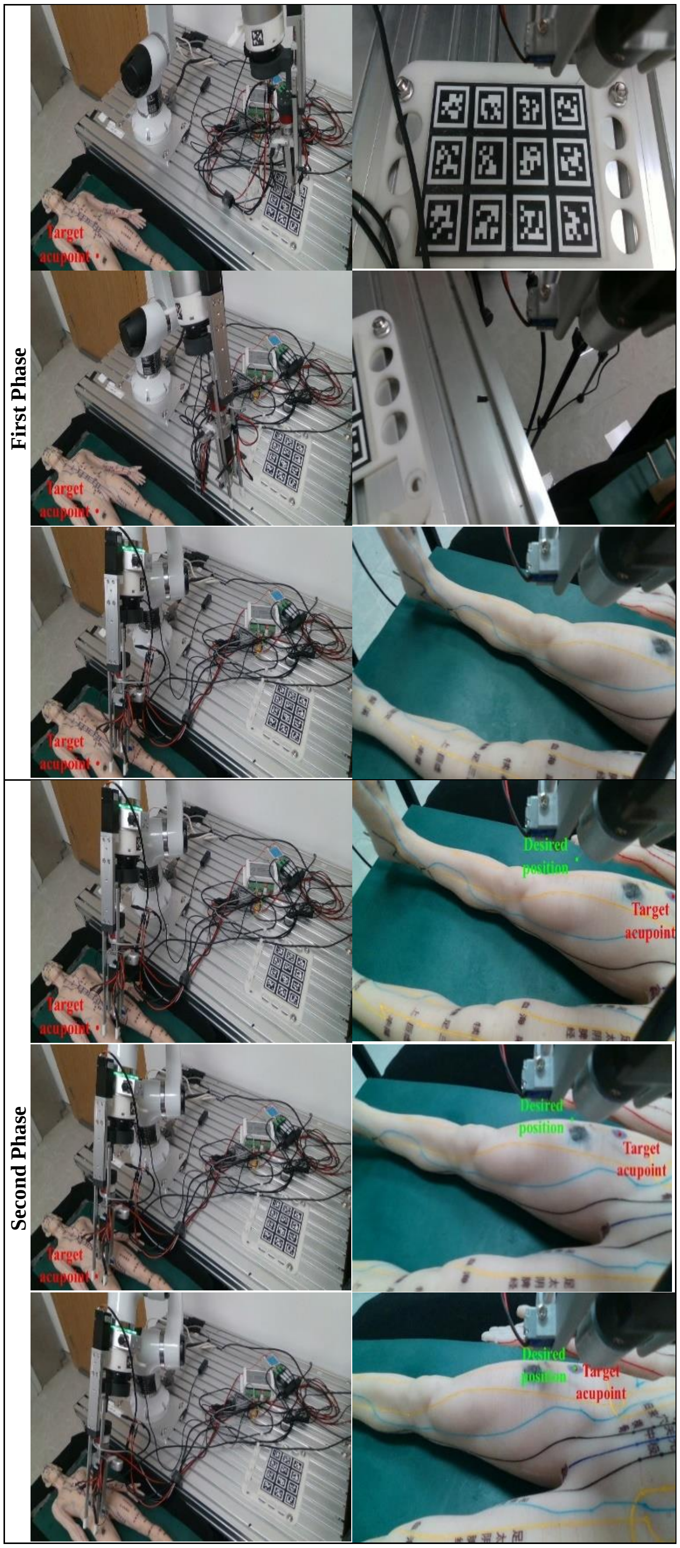
APPENDIX A

ACUPUNCTURE TRACKING AND ACUPUNCTURE CONTROL

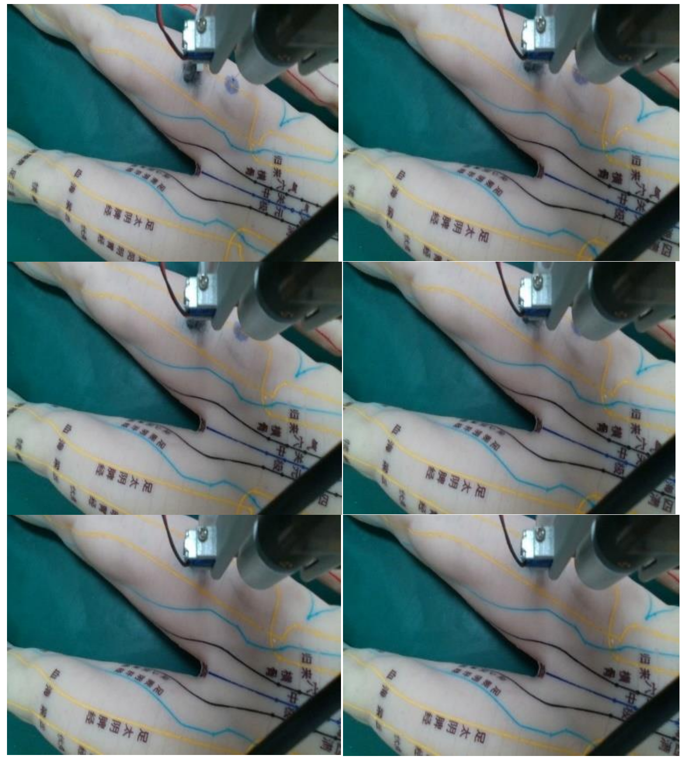
During the acupoint tracking process, based on the proposed tracking controller, the acupuncture camera images are shown in **FIG. A1**.



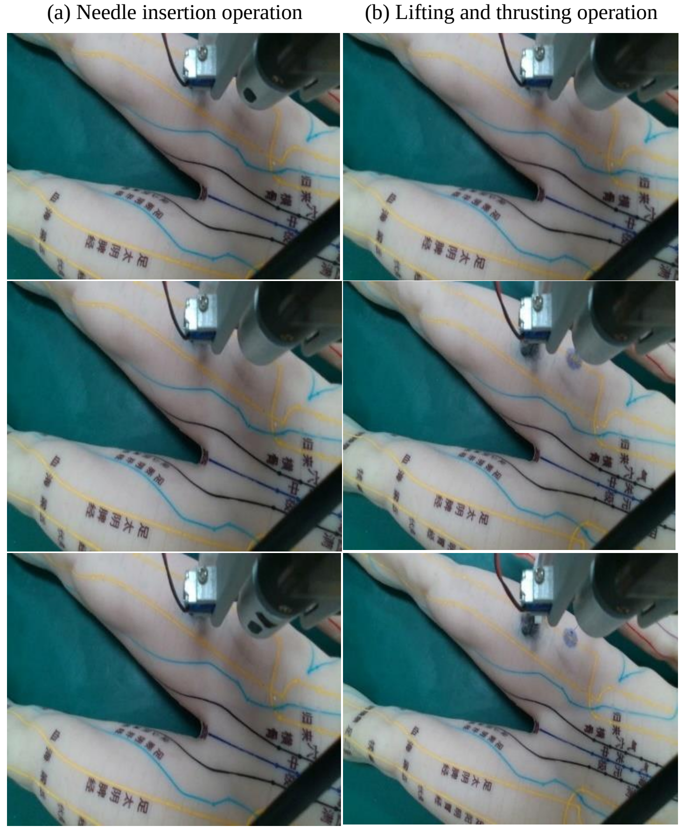
(a) Global image of acupuncture (b) Acupuncture eye-hand image

Fig. A1. Visual trajectory of acupoint tracking control

During the needle insertion process, based on the proposed adaptive impedance model-based compliant needle insertion controller and the classical PID method, the needle insertion images are shown in **FIG. A2**.



(a) Needle insertion operation (b) Lifting and thrusting operation



(c) Twisting operation (d) Needle withdrawal operation

Fig. A2. Needle insertion control process of the acupuncture robot

From **Fig. A2**, we can see that the acupuncture robot's needle insertion control process consists of four steps:

(1) Insertion: the insertion degree of freedom moves from the initial position d_0 to the insertion position d_1;

(2) Lifting and thrusting: the insertion degree of freedom moves from d_1 to d_2, then back to d_1, repeating multiple times;

(3) Twisting: the twisting degree of freedom moves from the initial position ϑ_0 to the twisting position ϑ_1, then back to ϑ_0, repeating multiple times;

(4) Withdrawal: the insertion degree of freedom moves from d_1 back to d_0.

APPENDIX B
PROOF OF STABILITY FOR THE HYBRID ACUPOINT TRACKING MODEL

Assuming that $d_{\text{track}} = \left\| \boldsymbol{s}_{\text{ap}} - \boldsymbol{s}_{\text{ap}}^* \right\|_2$ represents the acupoint tracking distance, based on the previous acupoint tracking model, the pseudoinverse-based solution model can be expressed as follows:

$$^{\text{end}}\boldsymbol{v}_{\text{track}} = \lambda_{\text{pos}} \frac{\boldsymbol{J}_{\text{camt}}^{\dagger}\left(\boldsymbol{s}_{\text{ap}}^* - \boldsymbol{s}_{\text{ap}}\right)}{\left\| \boldsymbol{J}_{\text{camt}}^{\dagger}\left(\boldsymbol{s}_{\text{ap}}^* - \boldsymbol{s}_{\text{ap}}\right)\right\|_2} \tag{B1}$$

$$\lambda_{\text{pos}} = K_{p,\text{pos}} d_{\text{track}} + K_{d,\text{pos}} \dot{d}_{\text{track}} \tag{B2}$$

where $\boldsymbol{J}_{\text{camt}}^{\dagger}$ represents the pseudoinverse of $\boldsymbol{J}_{\text{camt}}$, and $(K_{p,\text{pos}}, K_{d,\text{pos}})$ denotes the control parameter for acupoint tracking.

According to the pinhole imaging principle, the projection model of the acupuncture eye-in-hand camera can be expressed as:

$$\begin{bmatrix} u_{\text{ap}} \\ v_{\text{ap}} \\ 1 \end{bmatrix} = \frac{1}{^{\text{hec}}z_{\text{ap}}} \begin{bmatrix} f_u & 0 & u_0 \\ 0 & f_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} ^{\text{hec}}x_{\text{ap}} \\ ^{\text{hec}}y_{\text{ap}} \\ ^{\text{hec}}z_{\text{ap}} \end{bmatrix}. \tag{B3}$$

The linear velocity direction vector $\boldsymbol{e}_{\text{lin}}$ of the acupuncture eye-in-hand camera can be expressed as:

$$\boldsymbol{e}_{\text{lin}} = \begin{bmatrix} e_{\text{lin},x} \\ e_{\text{lin},y} \\ e_{\text{lin},z} \end{bmatrix} = \frac{^{\text{hec}}\boldsymbol{z}_{\text{hec}} \times {}^{\text{hec}}\hat{\boldsymbol{p}}_{\text{ap}} \times {}^{\text{hec}}\hat{\boldsymbol{p}}_{\text{ap}}}{\left\| ^{\text{hec}}\boldsymbol{z}_{\text{hec}} \times {}^{\text{hec}}\hat{\boldsymbol{p}}_{\text{ap}} \times {}^{\text{hec}}\hat{\boldsymbol{p}}_{\text{ap}} \right\|_2} \tag{B4}$$

where $^{\text{hec}}\boldsymbol{z}_{\text{hec}}$ represents the third column vector of the rotation matrix $^{\text{hec}}\boldsymbol{R}_{\text{hec}}$, $^{\text{hec}}\hat{\boldsymbol{p}}_{\text{ap}}$ represents the normalized position of the acupoint relative to the coordinate frame {hec}, $^{\text{hec}}\hat{\boldsymbol{p}}_{\text{ap}} = \left[\frac{^{\text{hec}}x_{\text{ap}}}{^{\text{hec}}z_{\text{ap}}}, \frac{^{\text{hec}}y_{\text{ap}}}{^{\text{hec}}z_{\text{ap}}}, 1 \right]^{\text{T}}$.

Assuming v_{track} represents the scalar linear velocity of the acupuncture eye-in-hand camera, then the linear velocity can be expressed as:

$$\boldsymbol{v}_{\text{hec}} = \begin{bmatrix} v_{\text{hec},x} \\ v_{\text{hec},y} \\ v_{\text{hec},z} \end{bmatrix} = v_{\text{track}} \boldsymbol{e}_{\text{lin}} = \frac{v_{\text{track}}}{l} \begin{bmatrix} ^{\text{hec}}x_{\text{ap}} \, ^{\text{hec}}z_{\text{ap}} \\ ^{\text{hec}}y_{\text{ap}} \, ^{\text{hec}}z_{\text{ap}} \\ ^{\text{hec}}x_{\text{ap}}^2 + {}^{\text{hec}}y_{\text{ap}}^2 \end{bmatrix} \tag{B5}$$

where l represents the velocity normalization scale, $l = \sqrt{^{\text{hec}}x_{\text{ap}}^2 + {}^{\text{hec}}y_{\text{ap}}^2} \left\| ^{\text{hec}}\boldsymbol{p}_{\text{ap}} \right\|_2$.

Assuming ($\omega_{\text{hec},x}$, $\omega_{\text{hec},y}$) represent the angular velocities of the acupuncture hand-eye camera around the X and Y axes respectively, and d_{hec} represents the distance between the origin of the acupuncture hand-eye camera coordinate system and the equivalent rotation center, then $\omega_{\text{hec},x} = -v_{\text{hec},y} / d_{\text{hec}}$, $\omega_{\text{hec},y} = v_{\text{hec},x} / d_{\text{hec}}$. Since the acupuncture hand-eye camera is moving and considering the slight movement of the acupoint, the velocity of the acupoint relative to the acupuncture hand-eye camera can be expressed as:

$$^{\text{hec}}\boldsymbol{v}_{\text{ap}} = \begin{bmatrix} ^{\text{hec}}\dot{x}_{\text{ap}} - v_{\text{hec},x} - {}^{\text{hec}}z_{\text{ap}}\omega_{\text{hec},y} \\ ^{\text{hec}}\dot{y}_{\text{ap}} - v_{\text{hec},y} + {}^{\text{hec}}z_{\text{ap}}\omega_{\text{hec},x} \\ ^{\text{hec}}\dot{z}_{\text{ap}} - v_{\text{hec},z} - {}^{\text{hec}}y_{\text{ap}}\omega_{\text{hec},x} + {}^{\text{hec}}x_{\text{ap}}\omega_{\text{hec},y} \end{bmatrix}$$

$$= \begin{bmatrix} ^{\text{hec}}\dot{x}_{\text{ap}} - \frac{v_{\text{track}}}{l} \, ^{\text{hec}}x_{\text{ap}} \, ^{\text{hec}}z_{\text{ap}} - \frac{v_{\text{track}}}{d_{\text{hec}} l} \, ^{\text{hec}}x_{\text{ap}} \, ^{\text{hec}}z_{\text{ap}}^2 \\ ^{\text{hec}}\dot{y}_{\text{ap}} - \frac{v_{\text{track}}}{l} \, ^{\text{hec}}y_{\text{ap}} \, ^{\text{hec}}z_{\text{ap}} - \frac{v_{\text{track}}}{d_{\text{hec}} l} \, ^{\text{hec}}y_{\text{ap}} \, ^{\text{hec}}z_{\text{ap}}^2 \\ ^{\text{hec}}\dot{z}_{\text{ap}} + \frac{v_{\text{track}}}{l}\left(^{\text{hec}}x_{\text{ap}}^2 + {}^{\text{hec}}y_{\text{ap}}^2 \right) + \frac{v_{\text{track}}}{d_{\text{hec}} l}\left(^{\text{hec}}x_{\text{ap}}^2 + {}^{\text{hec}}y_{\text{ap}}^2 \right) {}^{\text{hec}}z_{\text{ap}} \end{bmatrix} \tag{B6}$$

Differentiating both sides of Eq. (B3) and substituting Eq. (B6) into the differential result, we obtain:

$$\begin{aligned} \dot{u}_{\text{ap}} = \frac{f_u}{^{\text{hec}}z_{\text{ap}}^2} \Big[& ^{\text{hec}}\dot{x}_{\text{ap}} \, ^{\text{hec}}z_{\text{ap}} - {}^{\text{hec}}x_{\text{ap}} \, ^{\text{hec}}\dot{z}_{\text{ap}} \\ & - \frac{v_{\text{track}}}{l} \, ^{\text{hec}}x_{\text{ap}} \left(^{\text{hec}}x_{\text{ap}}^2 + {}^{\text{hec}}y_{\text{ap}}^2 + {}^{\text{hec}}z_{\text{ap}}^2 \right) \\ & - \frac{v_{\text{track}}}{d_{\text{hec}} l} \, ^{\text{hec}}x_{\text{ap}} \, ^{\text{hec}}z_{\text{ap}} \left(^{\text{hec}}x_{\text{ap}}^2 + {}^{\text{hec}}y_{\text{ap}}^2 + {}^{\text{hec}}z_{\text{ap}}^2 \right) \Big] \end{aligned} \tag{B7}$$

$$\begin{aligned} \dot{v}_{\text{tips}} = \frac{f_v}{^{\text{hec}}z_{\text{ap}}^2} \Big[& ^{\text{hec}}\dot{y}_{\text{ap}} \, ^{\text{hec}}z_{\text{ap}} - {}^{\text{hec}}y_{\text{ap}} \, ^{\text{hec}}\dot{z}_{\text{ap}} \\ & - \frac{v_{\text{track}}}{l} \, ^{\text{hec}}y_{\text{ap}} \left(^{\text{hec}}x_{\text{ap}}^2 + {}^{\text{hec}}y_{\text{ap}}^2 + {}^{\text{hec}}z_{\text{ap}}^2 \right) \\ & - \frac{v_{\text{track}}}{d_{\text{hec}} l} \, ^{\text{hec}}y_{\text{ap}} \, ^{\text{hec}}z_{\text{ap}} \left(^{\text{hec}}x_{\text{ap}}^2 + {}^{\text{hec}}y_{\text{ap}}^2 + {}^{\text{hec}}z_{\text{ap}}^2 \right) \Big] \end{aligned} \tag{B8}$$

Construct the Lyapunov function, which is expressed as the square of the acupoint tracking distance, that is:

$$L = \left\| \boldsymbol{s}_{\text{ap}} - \boldsymbol{s}_{\text{ap}}^* \right\|_2^2 = \left(u_{\text{ap}} - u_{\text{ap}}^* \right)^2 + \left(v_{\text{ap}} - v_{\text{ap}}^* \right)^2 \tag{B9}$$

Taking the derivative of both sides of Eq. (B9), yields:

$$\dot{L} = 2\left(u_{\text{ap}} - u_{\text{ap}}^* \right)\dot{u}_{\text{ap}} + 2\left(v_{\text{ap}} - v_{\text{ap}}^* \right)\dot{v}_{\text{ap}} \tag{B10}$$

Substituting Eqs. (B3), (B7), and (B8) into Eq. (B10), we obtain:

$$\begin{aligned}\dot{L} &= \frac{f_u^2\,{}^{\text{hec}}x_{\text{ap}}^2}{{}^{\text{hec}}z_{\text{ap}}^3}\left[\frac{{}^{\text{hec}}z_{\text{ap}}}{{}^{\text{hec}}x_{\text{ap}}}\,{}^{\text{hec}}\dot{x}_{\text{ap}} - {}^{\text{hec}}\dot{z}_{\text{ap}} - v_{\text{track}}\left(1+\frac{{}^{\text{hec}}z_{\text{ap}}}{d_{\text{hec}}}\right)\frac{\left\|{}^{\text{hec}}\boldsymbol{p}_{\text{ap}}\right\|_2}{\sqrt{{}^{\text{hec}}x_{\text{ap}}^2+{}^{\text{hec}}y_{\text{ap}}^2}}\right]\\ &\quad+\frac{f_v^2\,{}^{\text{hec}}y_{\text{ap}}^2}{{}^{\text{hec}}z_{\text{ap}}^3}\left[\frac{{}^{\text{hec}}z_{\text{ap}}}{{}^{\text{hec}}y_{\text{ap}}}\,{}^{\text{hec}}\dot{y}_{\text{ap}} - {}^{\text{hec}}\dot{z}_{\text{ap}} - v_{\text{track}}\left(1+\frac{{}^{\text{hec}}z_{\text{ap}}}{d_{\text{hec}}}\right)\frac{\left\|{}^{\text{hec}}\boldsymbol{p}_{\text{ap}}\right\|_2}{\sqrt{{}^{\text{hec}}x_{\text{ap}}^2+{}^{\text{hec}}y_{\text{ap}}^2}}\right]\end{aligned} \tag{B11}$$

When $d_{\text{hec}} \to +\infty$, Eq. (B11) can be simplified as:

$$\begin{aligned}\dot{L} &= \frac{f_u^2\,{}^{\text{hec}}x_{\text{ap}}^2}{{}^{\text{hec}}z_{\text{ap}}^3}\left[\frac{{}^{\text{hec}}z_{\text{ap}}}{{}^{\text{hec}}x_{\text{ap}}}\,{}^{\text{hec}}\dot{x}_{\text{ap}} - {}^{\text{hec}}\dot{z}_{\text{ap}} - \frac{v_{\text{track}}\left\|{}^{\text{hec}}\boldsymbol{p}_{\text{ap}}\right\|_2}{\sqrt{{}^{\text{hec}}x_{\text{ap}}^2+{}^{\text{hec}}y_{\text{ap}}^2}}\right]\\ &\quad+\frac{f_v^2\,{}^{\text{hec}}y_{\text{ap}}^2}{{}^{\text{hec}}z_{\text{ap}}^3}\left[\frac{{}^{\text{hec}}z_{\text{ap}}}{{}^{\text{hec}}y_{\text{ap}}}\,{}^{\text{hec}}\dot{y}_{\text{ap}} - {}^{\text{hec}}\dot{z}_{\text{ap}} - \frac{v_{\text{track}}\left\|{}^{\text{hec}}\boldsymbol{p}_{\text{ap}}\right\|_2}{\sqrt{{}^{\text{hec}}x_{\text{ap}}^2+{}^{\text{hec}}y_{\text{ap}}^2}}\right] \quad \text{(B12)}\\ &\le \frac{f_u^2\,{}^{\text{hec}}x_{\text{ap}}^2}{{}^{\text{hec}}z_{\text{ap}}^3}a_u + \frac{f_v^2\,{}^{\text{hec}}y_{\text{ap}}^2}{{}^{\text{hec}}z_{\text{ap}}^3}a_v\end{aligned}$$

$$a_u = \frac{{}^{\text{hec}}z_{\text{ap}}}{{}^{\text{hec}}x_{\text{ap}}}\,{}^{\text{hec}}\dot{x}_{\text{ap}} - {}^{\text{hec}}\dot{z}_{\text{ap}} - \frac{v_{\text{track}}\left\|{}^{\text{hec}}\boldsymbol{p}_{\text{ap}}\right\|_2}{\sqrt{{}^{\text{hec}}x_{\text{ap}}^2+{}^{\text{hec}}y_{\text{ap}}^2}} \tag{B13}$$

$$a_v = \frac{{}^{\text{hec}}z_{\text{ap}}}{{}^{\text{hec}}y_{\text{ap}}}\,{}^{\text{hec}}\dot{y}_{\text{ap}} - {}^{\text{hec}}\dot{z}_{\text{ap}} - \frac{v_{\text{track}}\left\|{}^{\text{hec}}\boldsymbol{p}_{\text{ap}}\right\|_2}{\sqrt{{}^{\text{hec}}x_{\text{ap}}^2+{}^{\text{hec}}y_{\text{ap}}^2}} \tag{B14}$$

Since it follows that:

$$a_u \le \max\left\{\frac{{}^{\text{hec}}z_{\text{ap}}}{\left|{}^{\text{hec}}x_{\text{ap}}\right|}v_{\text{track}}, v_{\text{track}}\right\} - \frac{v_{\text{track}}\left\|{}^{\text{hec}}\boldsymbol{p}_{\text{ap}}\right\|_2}{\sqrt{{}^{\text{hec}}x_{\text{ap}}^2+{}^{\text{hec}}y_{\text{ap}}^2}} \tag{B15}$$

$$a_v \le \max\left\{\frac{{}^{\text{hec}}z_{\text{ap}}}{\left|{}^{\text{hec}}y_{\text{ap}}\right|}v_{\text{track}}, v_{\text{track}}\right\} - \frac{v_{\text{track}}\left\|{}^{\text{hec}}\boldsymbol{p}_{\text{ap}}\right\|_2}{\sqrt{{}^{\text{hec}}x_{\text{ap}}^2+{}^{\text{hec}}y_{\text{ap}}^2}} \tag{B16}$$

Furthermore, it follows that:

$$\begin{cases} a_u \le v_{\text{track}} - \dfrac{v_{\text{track}}\left\|{}^{\text{hec}}\boldsymbol{p}_{\text{ap}}\right\|_2}{\sqrt{{}^{\text{hec}}x_{\text{ap}}^2+{}^{\text{hec}}y_{\text{ap}}^2}} \le 0, & \text{if } \dfrac{{}^{\text{hec}}z_{\text{ap}}}{\left|{}^{\text{hec}}x_{\text{ap}}\right|} \le 1\\ a_u \le \dfrac{{}^{\text{hec}}z_{\text{ap}}}{\left|{}^{\text{hec}}x_{\text{ap}}\right|}v_{\text{track}} - \dfrac{v_{\text{track}}\left\|{}^{\text{hec}}\boldsymbol{p}_{\text{ap}}\right\|_2}{\sqrt{{}^{\text{hec}}x_{\text{ap}}^2+{}^{\text{hec}}y_{\text{ap}}^2}} \le 0, & \text{if } \dfrac{{}^{\text{hec}}z_{\text{ap}}}{\left|{}^{\text{hec}}x_{\text{ap}}\right|} > 1 \end{cases} \tag{B17}$$

$$\begin{cases} a_v \le v_{\text{track}} - \dfrac{v_{\text{track}}\left\|{}^{\text{hec}}\boldsymbol{p}_{\text{ap}}\right\|_2}{\sqrt{{}^{\text{hec}}x_{\text{ap}}^2+{}^{\text{hec}}y_{\text{ap}}^2}} \le 0, & \text{if } \dfrac{{}^{\text{hec}}z_{\text{ap}}}{\left|{}^{\text{hec}}y_{\text{ap}}\right|} \le 1\\ a_v \le \dfrac{{}^{\text{hec}}z_{\text{ap}}}{\left|{}^{\text{hec}}y_{\text{ap}}\right|}v_{\text{track}} - \dfrac{v_{\text{track}}\left\|{}^{\text{hec}}\boldsymbol{p}_{\text{ap}}\right\|_2}{\sqrt{{}^{\text{hec}}x_{\text{ap}}^2+{}^{\text{hec}}y_{\text{ap}}^2}} \le 0, & \text{if } \dfrac{{}^{\text{hec}}z_{\text{ap}}}{\left|{}^{\text{hec}}y_{\text{ap}}\right|} > 1 \end{cases} \tag{B18}$$

Based on Eqs. (B12), (B17), and (B18), we obtain $\dot{L} \le 0$ .Therefore, the acupoint tracking error under the hybrid visual servoing model is convergent, which completes the proof.

APPENDIX C

PROOF OF STABILITY FOR THE COMPLIANT ACUPUNCTURE CONTROL MODEL

Since the 2-DOFs of the acupuncture device are decoupled, without loss of generality, this paper proves the stability of the compliant acupuncture control model for a single DOF. First, the stability of the impedance model based on tracking force error compensation is demonstrated, namely:

$$m_{\text{im}}\ddot{e} + b_{\text{im}}\dot{e} + k_{\text{im}}e = f_{\text{e}} + u\left(f_{\text{e}}, \dot{f}_{\text{e}}\right) \tag{C1}$$

$$u\left(f_{\text{e}}, \dot{f}_{\text{e}}\right) = \begin{cases} k_{\text{f}}\, f_{\text{e}}\dot{f}_{\text{e}}, & f_{\text{e}}\dot{f}_{\text{e}} > 0\\ 0, & f_{\text{e}}\dot{f}_{\text{e}} \le 0\end{cases} \tag{C2}$$

When the system tends toward stability, $e \to e_{\text{ss}}$, $\dot{e} \to 0$, $\ddot{e} \to 0$, $\ddot{e} \to 0$, $f_{\text{e}} \to f_{\text{ss}}$, $\dot{f}_{\text{e}} \to 0$. Considering any state near the equilibrium point of the system, let Δe and Δf_{e} represent the errors of the reference tracking position and tracking force error with respect to the system's equilibrium point, respectively. Then, we have:

$$\begin{gathered} e_{\text{as}} = e_{\text{ss}} + \Delta e,\ \dot{e}_{\text{as}} = \Delta\dot{e},\ \ddot{e}_{\text{as}} = \Delta\dot{e},\\ f_{\text{e,as}} = f_{\text{ss}} + \Delta f_{\text{e}},\ \dot{f}_{\text{e,as}} = \Delta\dot{f}_{\text{e}} \end{gathered} \tag{C3}$$

Combined Eq. (C1) with Eq. (C3), we obtain:

$$\begin{aligned} &m_{\text{im}}\Delta\ddot{e} + b_{\text{im}}\Delta\dot{e} + k_{\text{im}}\left(e_{\text{ss}} + \Delta e\right)\\ &= f_{\text{ss}} + \Delta f_{\text{e}} + u\left(f_{\text{ss}} + \Delta f_{\text{e}}, \Delta\dot{f}_{\text{e}}\right) \end{aligned} \tag{C4}$$

When $\left(f_{\text{ss}} + \Delta f_{\text{e}}\right)\Delta\dot{f}_{\text{e}} \le 0$, Eq. (C4) degenerates into the classical impedance control model, that is:

$$m_{\text{im}}\Delta\ddot{e} + b_{\text{im}}\Delta\dot{e} + k_{\text{im}}\left(e_{\text{ss}} + \Delta e\right) = f_{\text{ss}} + \Delta f_{\text{e}} \tag{C5}$$

Taking the Laplace transform of Eq. (C5), we obtain:

$$\left(m_{\text{im}}s^2 + b_{\text{im}}s + k_{\text{im}}\right)\Delta E\left(s\right) + k_{\text{im}}e_{\text{ss}} = \Delta F_{\text{e}}\left(s\right) + f_{\text{ss}} \tag{C6}$$

At this point, the system transfer function $G_1(s)$ can be expressed as:

$$G_1\left(s\right) = \frac{\Delta F_{\text{e}}\left(s\right)}{\Delta E\left(s\right)} = \left(m_{\text{im}}s^2 + b_{\text{im}}s + k_{\text{im}}\right) + \frac{k_{\text{im}}e_{\text{ss}} - f_{\text{ss}}}{\Delta E\left(s\right)} \tag{C7}$$

The stability of the classical impedance control model has already been proven, that is, all the poles of the transfer function $G_1(s)$ lie in the open left half-plane.

When $\left(f_{\text{ss}} + \Delta f_{\text{e}}\right)\Delta\dot{f}_{\text{e}} > 0$, Eq. (C4) can be simplified as follows:

$$\begin{aligned} &m_{\text{im}}\Delta\ddot{e} + b_{\text{im}}\Delta\dot{e} + k_{\text{im}}\left(e_{\text{ss}} + \Delta e\right)\\ &= f_{\text{ss}} + \Delta f_{\text{e}} + k_{\text{f}}\left(f_{\text{ss}} + \Delta f_{\text{e}}\right)\Delta\dot{f}_{\text{e}} \end{aligned} \tag{C8}$$

The term $\Delta f_{\text{e}}\Delta\dot{f}_{\text{e}}$ in Eq. (C8) is a higher-order term and can be approximated as 0, yielding:

$$m_{\text{im}}\Delta\ddot{e} + b_{\text{im}}\Delta\dot{e} + k_{\text{im}}\left(e_{\text{ss}} + \Delta e\right) = f_{\text{ss}} + \Delta f_{\text{e}} + k_{\text{f}}\, f_{\text{ss}}\Delta\dot{f}_{\text{e}} \tag{C9}$$

Taking the Laplace transform of Eq. (C9), we obtain:

$$\left(m_{\text{im}}s^2 + b_{\text{im}}s + k_{\text{im}}\right)\Delta E\left(s\right) + k_{\text{im}}e_{\text{ss}} = \left(k_{\text{f}}\, f_{\text{ss}}s + 1\right)\Delta F_{\text{e}}\left(s\right) + f_{\text{ss}} \tag{C10}$$

At this point, the system transfer function $G_2(s)$ can be expressed as:

$$G_2(s)=\frac{\Delta F_{\mathrm{e}}(s)}{\Delta E(s)}=\frac{G_1(s)}{k_{\mathrm{f}}f_{\mathrm{ss}}s+1}. \tag{C11}$$

Compared with the classical impedance control model, Eq. (C11) introduces an additional pole $s=-\frac{1}{k_{\mathrm{f}}f_{\mathrm{ss}}}$. When $k_{\mathrm{f}}>0$, this pole is always less than zero. Therefore, systems Eq. (C1) and (C2) are stable. Q.E.D.

Secondly, the stability of contact force tracking under the reference tracking position x_{r} is proven. When $z_{\mathrm{a}}\geq z_{\mathrm{ap}}$, the deformation model of the acupoint soft tissue in a single degree of freedom can be expressed as:

$$f_{\mathrm{ap}}=k_{\mathrm{ap}}\left(x_{\mathrm{a}}-x_{\mathrm{ap}}\right)+b_{\mathrm{ap}}\dot{x}_{\mathrm{a}} \tag{C12}$$

Under ideal conditions, $f_{\mathrm{a}}=f_{\mathrm{ap}}$, thus:

$$f_{\mathrm{e}}=f_{\mathrm{d}}-f_{\mathrm{a}}=f_{\mathrm{d}}-f_{\mathrm{ap}}=f_{\mathrm{d}}-k_{\mathrm{ap}}\left(x_{\mathrm{a}}-x_{\mathrm{ap}}\right)-b_{\mathrm{ap}}\dot{x}_{\mathrm{a}} \tag{C13}$$

According to the classical PID-based motor velocity loop control, it holds that:

$$\dot{x}_{\mathrm{a}}=v=\dot{x}_{\mathrm{d}}+k_{\mathrm{v}}(\hat{x}_{\mathrm{d}}-x_{\mathrm{a}}) \tag{C14}$$

Taking the Laplace transform of Eqs. (C13) and (C14), we obtain:

$$F_{\mathrm{e}}(s)+\left(b_{\mathrm{ap}}s+k_{\mathrm{ap}}\right)X_{\mathrm{a}}(s)=F_{\mathrm{d}}(s)+k_{\mathrm{ap}}X_{\mathrm{ap}}(s) \tag{C15}$$

$$\left(s+k_{\mathrm{v}}\right)X_{\mathrm{a}}(s)=\dot{X}_{\mathrm{d}}(s)+k_{\mathrm{v}}\hat{X}_{\mathrm{d}}(s) \tag{C16}$$

Combined Eq. (C15) with Eq. (C16), we get:

$$F_{\mathrm{e}}(s)+\left(b_{\mathrm{ap}}s+k_{\mathrm{ap}}\right)\frac{\dot{X}_{\mathrm{d}}(s)+k_{\mathrm{v}}\hat{X}_{\mathrm{d}}(s)}{s+k_{\mathrm{v}}}=F_{\mathrm{d}}(s)+k_{\mathrm{ap}}X_{\mathrm{ap}}(s) \tag{C17}$$

When $f_{\mathrm{e}}\dot{f}_{\mathrm{e}}\leq 0$, and since $e=\hat{x}_{\mathrm{d}}-x_{\mathrm{r}}$, taking the Laplace transform of Eq. (C1) yields:

$$\left(m_{\mathrm{im}}s^2+b_{\mathrm{im}}s+k_{\mathrm{im}}\right)\left[\hat{X}_{\mathrm{d}}(s)-X_{\mathrm{r}}(s)\right]=F_{\mathrm{e}}(s) \tag{C18}$$

Let $H_1(s)=\frac{1}{m_{\mathrm{im}}s^2+b_{\mathrm{im}}s+k_{\mathrm{im}}}$, then there is $\hat{X}_{\mathrm{d}}(s)=H_1(s)F_{\mathrm{e}}(s)+X_{\mathrm{r}}(s)$. Substituting $\hat{X}_{\mathrm{d}}(s)=H_1(s)F_{\mathrm{e}}(s)+X_{\mathrm{r}}(s)$ into Eq. (C18), yields:

$$\begin{aligned}&F_{\mathrm{e}}(s)+\left(b_{\mathrm{ap}}s+k_{\mathrm{ap}}\right)\frac{\dot{X}_{\mathrm{d}}(s)+k_{\mathrm{v}}\left[H_1(s)F_{\mathrm{e}}(s)+X_{\mathrm{r}}(s)\right]}{s+k_{\mathrm{v}}}\\&=F_{\mathrm{d}}(s)+k_{\mathrm{ap}}X_{\mathrm{ap}}(s)\end{aligned} \tag{C19}$$

Let $H_2(s)=\frac{b_{\mathrm{ap}}s+k_{\mathrm{ap}}}{s+k_{\mathrm{v}}}$, then Eq. (C19) can be simplified as:

$$\begin{aligned}&\left[k_{\mathrm{v}}H_1(s)H_2(s)+1\right]F_{\mathrm{e}}(s)+H_2(s)\dot{X}_{\mathrm{d}}(s)\\&+k_{\mathrm{v}}H_2(s)X_{\mathrm{r}}(s)=F_{\mathrm{d}}(s)+k_{\mathrm{ap}}X_{\mathrm{ap}}(s)\end{aligned} \tag{C20}$$

Since $\lim_{s\to 0}H_1(s)=\frac{1}{k_{\mathrm{im}}}$ and $\lim_{s\to 0}H_2(s)=\frac{k_{\mathrm{ap}}}{k_{\mathrm{v}}}$, the steady-state error of the contact force tracking can be expressed as:

$$\begin{aligned}f_{\mathrm{ss}}&=\lim_{s\to 0}sF_{\mathrm{e}}(s)\\&=\lim_{s\to 0}s\frac{F_{\mathrm{d}}(s)+k_{\mathrm{ap}}X_{\mathrm{ap}}(s)-H_2(s)\dot{X}_{\mathrm{d}}(s)-k_{\mathrm{v}}H_2(s)X_{\mathrm{r}}(s)}{k_{\mathrm{v}}H_1(s)H_2(s)+1}\\&=\frac{f_{\mathrm{d}}+k_{\mathrm{ap}}x_{\mathrm{ap}}-\frac{k_{\mathrm{ap}}}{k_{\mathrm{v}}}\dot{x}_{\mathrm{d}}-k_{\mathrm{v}}\frac{k_{\mathrm{ap}}}{k_{\mathrm{v}}}x_{\mathrm{r}}}{k_{\mathrm{v}}\frac{1}{k_{\mathrm{im}}}\frac{k_{\mathrm{ap}}}{k_{\mathrm{v}}}+1}\\&=\frac{k_{\mathrm{ap}}k_{\mathrm{im}}}{k_{\mathrm{ap}}+k_{\mathrm{im}}}\left(x_{\mathrm{ap}}+\frac{f_{\mathrm{d}}}{k_{\mathrm{ap}}}-\frac{\dot{x}_{\mathrm{d}}}{k_{\mathrm{v}}}-x_{\mathrm{r}}\right)\end{aligned} \tag{C21}$$

When $f_{\mathrm{e}}\dot{f}_{\mathrm{e}}>0$, taking the Laplace transform of Eq. (C1) yields:

$$\left(m_{\mathrm{im}}s^2+b_{\mathrm{im}}s+k_{\mathrm{im}}\right)\left[\hat{X}_{\mathrm{d}}(s)-X_{\mathrm{r}}(s)\right]=F_{\mathrm{e}}(s)+U(s) \tag{C22}$$

At this point, given $\hat{X}_{\mathrm{d}}(s)=H_1(s)F_{\mathrm{e}}(s)+H_1(s)U(s)+X_{\mathrm{r}}(s)$, substituting it into Eq. (C17) yields:

$$\begin{aligned}&\left[k_{\mathrm{v}}H_1(s)H_2(s)+1\right]F_{\mathrm{e}}(s)+k_{\mathrm{v}}H_1(s)H_2(s)U(s)\\&=F_{\mathrm{d}}(s)+k_{\mathrm{ap}}X_{\mathrm{ap}}(s)-H_2(s)\dot{X}_{\mathrm{d}}(s)-k_{\mathrm{v}}H_2(s)X_{\mathrm{r}}(s)\end{aligned} \tag{C23}$$

Similarly, the steady-state error of contact force tracking can be expressed as:

$$\begin{aligned}f_{\mathrm{ss}}&=\lim_{s\to 0}sF_{\mathrm{e}}(s)\\&=\frac{k_{\mathrm{ap}}k_{\mathrm{im}}}{k_{\mathrm{ap}}+k_{\mathrm{im}}}\left(x_{\mathrm{ap}}+\frac{f_{\mathrm{d}}}{k_{\mathrm{ap}}}-\frac{\dot{x}_{\mathrm{d}}}{k_{\mathrm{v}}}-x_{\mathrm{r}}\right)-\lim_{s\to 0}s\frac{k_{\mathrm{v}}H_1(s)H_2(s)U(s)}{k_{\mathrm{v}}H_1(s)H_2(s)+1}\\&=\frac{k_{\mathrm{ap}}k_{\mathrm{im}}}{k_{\mathrm{ap}}+k_{\mathrm{im}}}\left(x_{\mathrm{ap}}+\frac{f_{\mathrm{d}}}{k_{\mathrm{ap}}}-\frac{\dot{x}_{\mathrm{d}}}{k_{\mathrm{v}}}-\frac{u}{k_{\mathrm{im}}}-x_{\mathrm{r}}\right)\end{aligned} \tag{C24}$$

The reference tracking position can be denoted as $x_{\mathrm{r}}=x_{\mathrm{ap}}+\frac{f_{\mathrm{d}}}{k_{\mathrm{ap}}}-\frac{\dot{x}_{\mathrm{d}}}{k_{\mathrm{v}}}-\frac{u\left(f_{\mathrm{e}},\dot{f}_{\mathrm{e}}\right)}{k_{\mathrm{im}}}$. Substituting it into Eqs. (C21) and (C24) shows that $f_{\mathrm{ss}}=0$ always holds, thus completing the proof.

APPENDIX D

ACUPOINT POSE RECONSTRUCTION METHOD

The position and normal vector of the acupoint can be reconstructed using monocular vision; however, the solution is not unique. The quadratic form of the elliptical (i.e., Eq. (4)) can be expressed as:

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix}^{\mathrm{T}} \underbrace{\begin{bmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{bmatrix}}_{\boldsymbol{E}} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = 0 \tag{D1}$$

where $\boldsymbol{E}$ denotes the parameter matrix of the quadratic form of an elliptical equation.

Assume that $\boldsymbol{p}=[x,y,z]^{\mathrm{T}}$ represents the Cartesian coordinates in the acupuncture camera space corresponding to pixel coordinates $\boldsymbol{s}$.

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \frac{1}{z} \underbrace{\begin{bmatrix} f_u & 0 & u_0 \\ 0 & f_v & v_0 \\ 0 & 0 & 1 \end{bmatrix}}_{\boldsymbol{K}} \boldsymbol{p} \tag{D2}$$

where (f_u, f_v, u_0, v_0) denotes the intrinsic parameters of the acupuncture camera, which constitute the intrinsic matrix $\boldsymbol{K}$.

Combined Eq. (D1) with Eq. (D2), the quadratic form equation of the oblique circular cone in the acupuncture camera space can be expressed as:

$$\boldsymbol{p}^{\mathrm{T}} \boldsymbol{H} \boldsymbol{p} = 0 \tag{D3}$$

where $\boldsymbol{H}$ denotes the parameter matrix of the oblique conical quadric equation, $\boldsymbol{H}=\boldsymbol{K}^{\mathrm{T}}\boldsymbol{E}\boldsymbol{K}$.

Assuming that λ_1, λ_2, and λ_3 are the three eigenvalues of $\boldsymbol{H}$, satisfying $\lambda_1 \geq \lambda_2 > 0 > \lambda_3$, and $\boldsymbol{q}_1$, $\boldsymbol{q}_2$, and $\boldsymbol{q}_3$ are the corresponding unit eigenvectors. Therefore, in the acupuncture camera space, the normalized transformation matrix of the oblique conical quadric can be expressed as $\boldsymbol{P} = [\boldsymbol{o}_1, \boldsymbol{o}_2, \boldsymbol{o}_3]$, where $\boldsymbol{o}_3 = (-1)^{1+\mathrm{sgn}(\boldsymbol{q}_{3,z})} \boldsymbol{q}_3$, $q_{3,z}$ denotes the third element of $\boldsymbol{q}_3$, sgn(•) is the sign function, $\boldsymbol{o}_2=\boldsymbol{q}_2$, $\boldsymbol{o}_1=\boldsymbol{o}_2 \times \boldsymbol{o}_3$.

Accordingly, the normal vector and position of the acupoint can be expressed as:

$$(\boldsymbol{n}_{\mathrm{ap},1}, \boldsymbol{n}_{\mathrm{ap},2}) = \boldsymbol{P} \left[\pm\sqrt{\frac{\lambda_1 - \lambda_2}{\lambda_1 - \lambda_3}}, 0, -\sqrt{\frac{\lambda_2 - \lambda_3}{\lambda_1 - \lambda_3}} \right]^{\mathrm{T}} \tag{D4}$$

$$(\boldsymbol{p}_{\mathrm{ap},1}, \boldsymbol{p}_{\mathrm{ap},2}) = r_{\mathrm{ap}} \underbrace{\boldsymbol{P} \left[\pm\sqrt{-\frac{\lambda_3(\lambda_1 - \lambda_2)}{\lambda_1(\lambda_1 - \lambda_3)}}, 0, \sqrt{-\frac{\lambda_1(\lambda_2 - \lambda_3)}{\lambda_3(\lambda_1 - \lambda_3)}} \right]^{\mathrm{T}}}_{(\boldsymbol{k}_{\mathrm{ap},1}, \boldsymbol{k}_{\mathrm{ap},2})} \tag{D5}$$

where r_{ap} represents the radius of the acupoint.

Algorithm D1: Acupoint pose reconstruction method

Input: $\boldsymbol{E}^{\mathrm{hec}}$, $\boldsymbol{E}^{\mathrm{gc}}$, $\boldsymbol{K}^{\mathrm{hec}}$ and $\boldsymbol{K}^{\mathrm{gc}}$

Ouput: ${}^{\mathrm{hec}}\boldsymbol{n}_{\mathrm{ap}}^{\mathrm{hec}}$, ${}^{\mathrm{gc}}\boldsymbol{n}_{\mathrm{ap}}^{\mathrm{gc}}$, ${}^{\mathrm{hec}}\boldsymbol{p}_{\mathrm{ap}}^{\mathrm{hec}}$ and ${}^{\mathrm{gc}}\boldsymbol{p}_{\mathrm{ap}}^{\mathrm{gc}}$

Calculate $\boldsymbol{H}^{\mathrm{hec}}=(\boldsymbol{K}^{\mathrm{hec}})^{\mathrm{T}}\boldsymbol{E}^{\mathrm{hec}}\boldsymbol{K}^{\mathrm{hec}}$ and $\boldsymbol{H}^{\mathrm{gc}}=(\boldsymbol{K}^{\mathrm{gc}})^{\mathrm{T}}\boldsymbol{E}^{\mathrm{gc}}\boldsymbol{K}^{\mathrm{gc}}$ according to Eqs. (D1) to (D3).

Obtain $\boldsymbol{P}^{\mathrm{hec}}$ based on $\boldsymbol{H}^{\mathrm{hec}}$, then calculate $({}^{\mathrm{hec}}\boldsymbol{n}_{\mathrm{ap},1}^{\mathrm{hec}}, {}^{\mathrm{hec}}\boldsymbol{n}_{\mathrm{ap},2}^{\mathrm{hec}})$ and $({}^{\mathrm{hec}}\boldsymbol{k}_{\mathrm{ap},1}^{\mathrm{hec}}, {}^{\mathrm{hec}}\boldsymbol{k}_{\mathrm{ap},2}^{\mathrm{hec}})$ according to Eqs. (D4) and (D5).

Obtain $\boldsymbol{P}^{\mathrm{gc}}$ based on $\boldsymbol{H}^{\mathrm{gc}}$, then calculate $({}^{\mathrm{gc}}\boldsymbol{n}_{\mathrm{ap},1}^{\mathrm{gc}}, {}^{\mathrm{gc}}\boldsymbol{n}_{\mathrm{ap},2}^{\mathrm{gc}})$ and $({}^{\mathrm{gc}}\boldsymbol{k}_{\mathrm{ap},1}^{\mathrm{gc}}, {}^{\mathrm{gc}}\boldsymbol{k}_{\mathrm{ap},2}^{\mathrm{gc}})$ according to Eqs. (D4) and (D5).

Calculate $\psi_{i,j}$ (i=1, 2 and j=1,2) according to Eq. (16), and determine the indices $i_{\min}$ and $j_{\min}$ corresponding to the minimum value.

${}^{\mathrm{hec}}\boldsymbol{n}_{\mathrm{ap}}^{\mathrm{hec}} = {}^{\mathrm{hec}}\boldsymbol{n}_{\mathrm{ap},i_{\min}}^{\mathrm{hec}}$, ${}^{\mathrm{hec}}\boldsymbol{k}_{\mathrm{ap}}^{\mathrm{hec}} = {}^{\mathrm{hec}}\boldsymbol{k}_{\mathrm{ap},i_{\min}}^{\mathrm{hec}}$, ${}^{\mathrm{gc}}\boldsymbol{n}_{\mathrm{ap}}^{\mathrm{gc}} = {}^{\mathrm{gc}}\boldsymbol{n}_{\mathrm{ap},j_{\min}}^{\mathrm{gc}}$, ${}^{\mathrm{gc}}\boldsymbol{k}_{\mathrm{ap}}^{\mathrm{gc}} = {}^{\mathrm{gc}}\boldsymbol{k}_{\mathrm{ap},j_{\min}}^{\mathrm{gc}}$

Calculate r_{ap} according to Eq. (18), and then calculate ${}^{\mathrm{hec}}\boldsymbol{p}_{\mathrm{ap}}^{\mathrm{hec}} = r_{\mathrm{ap}}\, {}^{\mathrm{hec}}\boldsymbol{k}_{\mathrm{ap}}^{\mathrm{hec}}$ and ${}^{\mathrm{gc}}\boldsymbol{p}_{\mathrm{ap}}^{\mathrm{gc}} = r_{\mathrm{ap}}\, {}^{\mathrm{gc}}\boldsymbol{k}_{\mathrm{ap}}^{\mathrm{gc}}$.

APPENDIX E

SYSTEM CALIBRATION EXPERIMENT OF THE ACUPUNCTURE ROBOT

To verify the correctness and effectiveness of the proposed acupoint pose estimation method, we simulated the generation of six acupoint datasets. Their projections on the images are shown in **Fig. E1**, where AP1 to AP6 represent the six acupoints, respectively. In the eye-in-hand images, the visible areas of AP1 and AP2 were less than 50%, the visible areas of AP3 and AP4 were greater than 50%, and AP5 and AP6 were fully visible. In the global image, all the acupoints were fully visible. Then, pixel points were sampled from the elliptical contours of the six acupoints, with zero-mean Gaussian noise of variance 4 added to each point.

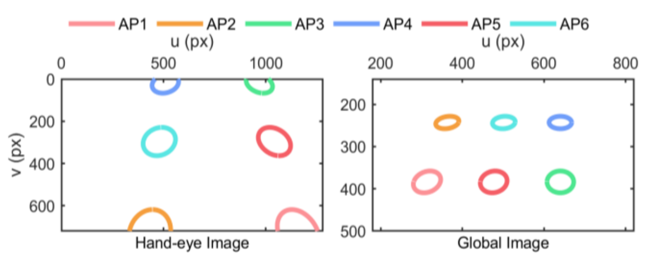


Fig. E1. Projection image of acupoint markers

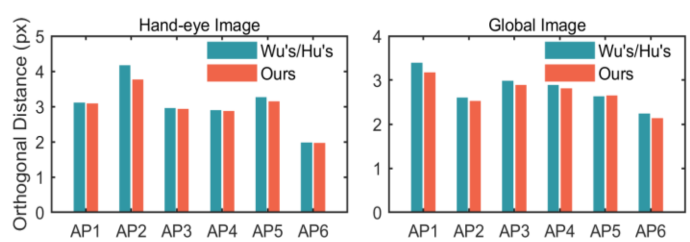


Fig. E2. Bar chart of orthogonal distances for elliptical contour fitting of acupoints

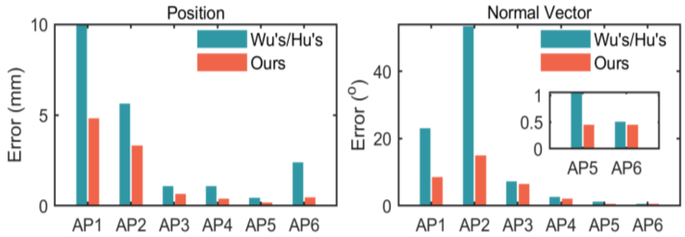


Fig. E3. Bar chart of pose estimation errors for acupoints

Finally, the position and normal vector of the acupoints were reconstructed based on the spatial circle method. Xu *et al.* [34] used a monocular camera to estimate the 2D planar displacement of circular markers, which cannot be applied to acupoint pose estimation. Therefore, this study compares the proposed method with the optimization-free methods presented by Wu *et al.* [35] and Hu *et al.* [36]. The comparison results of acupoint elliptical contour fitting and pose estimation are shown in **Fig. E2** and **Fig. E3**, respectively. For AP1 and AP2, which have smaller visible areas, optimization-free methods use arithmetic distance-based fitting algorithms, which easily cause large errors or even failures in pose estimation, adversely affecting the subsequent acupoint tracking performance of the acupuncture robot. The proposed method optimizes the acupoint elliptical contour parameters based on the approximate geometric distance, thereby reducing the orthogonal distance of fitting and errors in pose estimation. For the other acupoints, both methods achieved relatively high-accuracy pose estimation results. Notably, owing to the joint optimization of the acupoint position and normal vector, the pose estimation accuracy of the proposed method surpassed that of optimization-free methods. During actual acupoint tracking control, as the elliptical contour of the acupoint changes from partially visible to fully visible, the proposed method ultimately satisfies the requirement for high-precision pose estimation.

APPENDIX F

SYSTEM CALIBRATION EXPERIMENT OF THE ACUPUNCTURE ROBOT

In the experimental system, the D-H parameters of the 6-DOF acupuncture manipulator are shown in **Tab. F1**. The key parameters of the acupuncture global and eye-hand cameras are shown in **Tab. F2.** The parameters of the acupuncture force/torque sensor are shown in **Tab. F3.**

Tab. F1. The kinematic parameters of the acupuncture manipulator

k	α_k (°)	a_k (mm)	d_k (mm)	$\theta_{\mathrm{off},k}$ (°)
1	90.0	0.0	192.5	0.0
2	180.0	266.0	0.0	90.0
3	90.0	0.0	0.0	90.0
4	-90.0	0.0	324.0	0.0
5	90.0	0.0	0.0	0.0
6	0.0	0.0	155.0	0.0

Tab. F2. The key parameters of the acupuncture global and eye-hand cameras

Parameters	RGB module	Depth module
Frame resolution (Horizontal×vertical)	1920×1080	1280×720
Field of view (Horizontal×vertical)	69°×42°	65°×40°
Frame rate	30fps	90fps

Tab. F3. The key parameters of the acupuncture force/torque sensor

Parameters	Pressure sensor	Torque sensor
type	SIMBATOUCH SBT650	ARIZON AR-AT10S
Range	[−50, 50] N	[−0.5, 0.5] Nm
Accuracy	0.1% F.S	0.1% F.S

The acupuncture needle tip is fixed at the same position in space, and the end-effector pose of the acupuncture robotic arm is randomly adjusted to collect 50 sets of experimental data samples. Each set of samples includes corresponding robotic arm joint angles, global images, and eye-in-hand images. Unlike numerical simulations, system calibration in the real environment also requires reconstructing measurement data based on the collected joint angles and acupuncture images, i.e., homogeneous transformation matrices. Specifically, the forward kinematics model of the acupuncture manipulator is used to obtain $\boldsymbol{T}_A$, while $\boldsymbol{T}_B$ and $\boldsymbol{T}_C$ are reconstructed based on AprilTag detection and the Perspective-*n*-Point algorithm. Therefore, the noise mainly originates from data acquisition errors and measurement reconstruction errors.

Assuming that $\boldsymbol{K}_{\mathrm{in,hec}}$ and $\boldsymbol{K}_{\mathrm{in,gc}}$ represent the intrinsic parameter matrices of the acupuncture eye-in-hand camera and the global camera, respectively, then:

$$\boldsymbol{K}_{\mathrm{in,hec}} = \begin{bmatrix} 916.59 & 0 & 639.71 \\ 0 & 920.74 & 353.50 \\ 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{K}_{\mathrm{in,gc}} = \begin{bmatrix} 898.65 & 0 & 648.37 \\ 0 & 896.32 & 368.63 \\ 0 & 0 & 1 \end{bmatrix} \tag{F1}$$

Furthermore, the extrinsic calibration results of the acupuncture robot can be expressed as:

$${}^{0}\hat{\boldsymbol{T}}_{\mathrm{gc}} = \begin{bmatrix} -0.0530 & -0.4425 & 0.8952 & -350.1839 \\ -0.9982 & -0.0006 & -0.0594 & 437.2193 \\ 0.0268 & -0.8968 & -0.4416 & 315.3470 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{F2}$$

$${}^{\mathrm{e}}\hat{\boldsymbol{T}}_{\mathrm{lat}} = \begin{bmatrix} -0.0036 & 0.9999 & 0.0131 & 4.6525 \\ 0.0249 & -0.0130 & 0.9996 & 38.6233 \\ 0.9997 & 0.0039 & -0.0249 & 10.7463 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{F3}$$

$${}^{\mathrm{e}}\hat{\boldsymbol{T}}_{\mathrm{hec}} = \begin{bmatrix} -0.0131 & -0.9999 & 0.0077 & 64.6258 \\ 0.9999 & -0.0130 & 0.0105 & -37.3525 \\ -0.0104 & 0.0078 & 0.9999 & 31.9667 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{F4}$$

$$ {}^{\mathrm{e}}\hat{\boldsymbol{t}}_{\mathrm{an}} = \begin{bmatrix} -0.0002 \\ -0.0007 \\ 233.9964 \end{bmatrix} \tag{F5} $$